$$\bar{X} = \hat{\mu}, s = \hat{\sigma}, SE_{\bar{X}} = s \div \sqrt{n}$$

$$n = \left(z_{\alpha/2}\sigma \div SE\right)^2$$

C.I. of μ: $\bar{X} \pm t_{\alpha/2,n-1} SE_{\bar{X}}$

one

1 set μ, H, α; 2 check CLT / norm; 3 calc $\bar{X}$, s; then T

$$T = \frac{\bar{X} - c}{s/\sqrt{n}} \sim t(n-1)$$

$$\text{Reject } H_0 \text{ if: } \begin{cases} H_1 : u \neq c & |T| > t_{\alpha/2,n-1} \\ & p = 2P(t > |T|) \\ H_1 : u > c & T > t_{\alpha,n-1} \\ & p = P(t > T) \\ H_1 : u < c & T < -t_{\alpha,n-1} \\ & p = P(t < T) \end{cases}$$

also rej. H_0 if p-value $< \alpha$

forgetful

$$s^2 = \frac{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}{n-1}$$

2 independent

$$\sigma_1 = \sigma_2 \text{ if } \frac{max(s_1, s_2)}{min(s_1, s_2)} < 2$$

$$\sigma_1 = \sigma_2 : \begin{cases} T & = \frac{\bar{X}_1 - \bar{X}_2}{SE_{\text{pooled}}} \\ df^* & = n_1 + n_2 - 2 \end{cases}$$

$$\sigma_1 \neq \sigma_2 : \begin{cases} T & = \frac{\bar{X}_1 - \bar{X}_2}{SE_{\text{unpooled}}} \\ df^* & = min(n_1 - 1, n_2 - 1) \end{cases}$$

$$SE_{u..} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \; SE_{p..} = \sqrt{S_{p..}^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$S_{p..}^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\text{Reject } H_0 \text{ if: } \begin{cases} H_1 : \mu_1 - \mu_2 \neq 0 & |T| > t_{\alpha/2,df} \\ H_1 : \mu_1 - \mu_2 > 0 & |T| > t_{\alpha,df} \\ H_1 : \mu_1 - \mu_2 < 0 & |T| < -t_{\alpha,df} \end{cases}$$

indp C.I.: $(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2,df} SE_?$

accept H_0 if has zero

2 paired

$$\text{Paired: } \begin{cases} H_0 : \text{use } \mu_d \\ d_i = x_{1i} - x_{2i} \forall i = 1, ..., n \\ \text{compute } \bar{d}, s_d, SE_{\bar{d}} \\ SE_{\bar{d}} = s_d/\sqrt{n} \\ T = \frac{\bar{d}}{SE_{\bar{d}}} \sim t(n-1) \end{cases}$$

$$\text{Reject } H_0 \text{ if: } \begin{cases} H_1 : \mu_d \neq 0 & |T| > t_{\alpha/2, n-1} \\ H_1 : \mu_d > 0 & T > t_{\alpha, n-1} \\ H_1 : \mu_d < 0 & T < -t_{\alpha, n-1} \end{cases}$$

$$\text{pair C.I.: } \bar{d} \pm t_{\alpha/2, n-1} SE_{\bar{d}}$$

accept H_0 if has zero

pop param

$$\hat{p}\text{: } \begin{cases} \hat{p} & = \frac{X}{n} \\ \hat{p} & \sim N(p, \sqrt{\frac{p(1-p)}{n}}) \end{cases}$$

X: # of outcms w/ a char.

1 set p, H, α; 2 check np and $n(1-p) \geq 5$; 3 calc $\hat{p}$; then Z

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \sim N(0, 1)$$

$p = p_0$ under H_0

$$\text{Reject } H_0 \text{ if: } \begin{cases} H_1 : p \neq p_0 & |Z| > z_{\alpha/2} \\ & 2P(z > |Z|) < \alpha \\ H_1 : p > p_0 & Z > Z_\alpha \\ & P(z > Z) < \alpha \\ H_1 : p < p_0 & Z < -Z_\alpha \\ & P(z < Z) < \alpha \end{cases}$$

$$\text{C.I. for } p\text{: } \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$